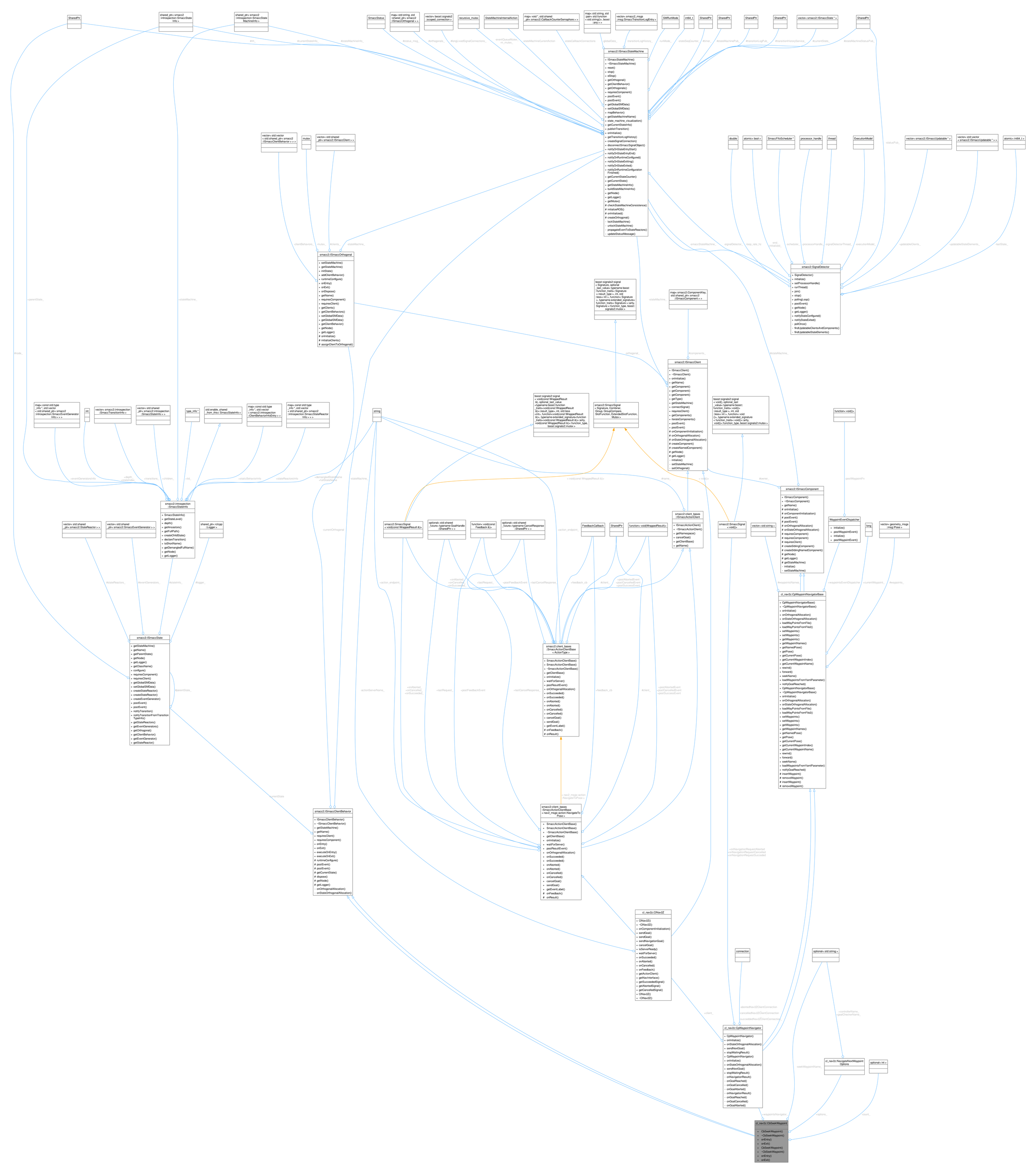
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
vector< boost::signals2 ::scoped_connection >
recursive_mutex
StateMachineInternalAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
SharedPtr
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
SharedPtr
#sh_
#currentStateInfo_
#stateMachineInfo_
#status_msg_
#orthogonals_
#longLivedSignalConnections_
-eventQueueMutex _m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#timer_
#stateMachinePub_
#transitionLogPub_
#transitionHistoryService_
#currentState_
#stateMachineStatusPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ mapBehavior()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
mutex
vector< std::shared _ptr< smacc2::ISmaccClient > >
-clientBehaviors_
-mutex_
#clients_
-stateMachine_
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-statusPub_
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
-smaccStateMachine_
-signalDetector_
-loop_rate_hz
-end_ initialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-lastState_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
#parentState_
#stateMachine_
#node_
-orthogonal_
#components_
#stateMachine_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onOrthogonalAllocation()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < void(const WrappedResult &), optional_last_value <typename boost::function _traits<void(const WrappedResult &)>::result_type>, int, std::less <int>, function<void(const WrappedResult &)>, typename extended_signature<function _traits<void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()>::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
function< void()>
map< const std::type _info *, std::vector < smacc2::introspection ::SmaccEventGenerator Info > >
int
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
std::enable_shared from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
string
+eventGeneratorsInfo_
+depth_ +stateIndex_
+transitions_
+children_
+id_
+staticBehaviorInfo
+stateReactorsInfo
+demangledStateName_ +fullStateName_
-stateMachine_
< void(const WrappedResult &)>
#name_
< void()>
#owner_
#postWaypointFn
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::SmaccSignal < void(const WrappedResult &)>
optional< std::shared _future< typename GoalHandle ::SharedPtr > >
function< void(const Feedback &)>
optional< std::shared _future< typename CancelResponse ::SharedPtr > >
-action_endpoint_
FeedbackCallback
SharedPtr
function< void(WrappedResult)>
smacc2::client_bases ::SmaccActionClient
+ ISmaccActionClient()
+ ~ISmaccActionClient()
+ getNamespace()
+ cancelGoal()
+ getClientBase()
+ getName()
smacc2::SmaccSignal < void()>
vector< std::string >
WaypointEventDispatcher
+ initialize()
+ postWaypointEvent()
+ initialize()
+ postWaypointEvent()
long
vector< geometry_msgs ::msg::Pose >
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onOrthogonalAllocation()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
#waypointsNames_
+waypointsEventDispatcher
+currentWaypoint_
#waypoints_
cl_nav2z::ClpWaypointNavigatorBase
+ ClpWaypointNavigatorBase()
+ ~ClpWaypointNavigatorBase()
+ oninitialize()
+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
+ ClpWaypointNavigatorBase()
+ ~ClpWaypointNavigatorBase()
+ oninitialize()
+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
# insertWaypoint()
# removeWaypoint()
# insertWaypoint()
# removeWaypoint()
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
shared_ptr< rclcpp ::Logger >
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
smacc2::client_bases ::SmaccActionClientBase < ActionType >
+ SmaccActionClientBase()
+ SmaccActionClientBase()
+ ~SmaccActionClientBase()
+ getClientBase()
+ onInitialize()
+ waitForServer()
+ postResultEvent()
+ onOrthogonalAllocation()
+ onSucceeded()
+ onSucceeded()
+ onAborted()
+ onAborted()
+ onCancelled()
+ onCancelled()
+ cancelGoal()
+ sendGoal()
+ getEventLabel()
# onFeedback()
# onResult()
#stateReactors_
#eventGenerators_
#stateInfo_
#logger_
-currentOrthogonal
-action_endpoint_
+onAborted_ +onCancelled_ +onSucceeded_
+lastRequest_
+postFeedbackEvent
+lastCancelResponse_
+feedback_cb
#client_
+postAbortedEvent +postCancelledEvent +postSuccessEvent
-actionServerName_
-currentState
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onOrthogonalAllocation()
- onStateOrthogonalAllocation()
smacc2::client_bases ::SmaccActionClientBase < nav2_msgs::action::NavigateTo Pose >
+ SmaccActionClientBase()
+ SmaccActionClientBase()
+ ~SmaccActionClientBase()
+ getClientBase()
+ onInitialize()
+ waitForServer()
+ postResultEvent()
+ onOrthogonalAllocation()
+ onSucceeded()
+ onSucceeded()
+ onAborted()
+ onAborted()
+ onCancelled()
+ onCancelled()
+ cancelGoal()
+ sendGoal()
+ getEventLabel()
# onFeedback()
# onResult()
< nav2_msgs::action ::NavigateToPose >
+onNavigationRequestAborted +onNavigationRequestCancelled +onNavigationRequestSucceeded
cl_nav2z::ClNav2Z
+ ClNav2Z()
+ ~ClNav2Z()
+ onComponentInitialization()
+ sendGoal()
+ sendGoal()
+ sendNavigationGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
+ getNavInterface()
+ getSucceededSignal()
+ getAbortedSignal()
+ getCancelledSignal()
+ ClNav2Z()
+ ~ClNav2Z()
connection
optional< std::string >
+client_
-abortedNav2ZClientConnection
-cancelledNav2ZClientConnection
-succeededNav2ZClientConnection
+controllerName_ +goalCheckerName_
cl_nav2z::ClpWaypointNavigator
+ ClpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
+ ClpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
cl_nav2z::NavigateNextWaypoint Options
optional< int >
-seekWaypointName_
+waypointsNavigator_
+options_
+count_
cl_nav2z::CbSeekWaypoint
+ CbSeekWaypoint()
+ ~CbSeekWaypoint()
+ onEntry()
+ onExit()
+ CbSeekWaypoint()
+ ~CbSeekWaypoint()
+ onEntry()
+ onExit()